



Blade Series — Lecture 11

Topper Notes — Questions + Detailed Explanation | 24 Topics Covered

 Ecology ·  Mughal History ·  Biology ·  Polity ·  Geography ·  Chemistry ·  Economics



→ Join Telegram: t.me/Exam_Zen

① Ecology — Biodiversity Hotspots

Q1. The Western Ghats are a recognized biodiversity hotspot. Which of the following is currently considered the MOST SIGNIFICANT threat to its endemic species, beyond general habitat loss?

- ☐ A Infrastructure development projects
- ☒ B Introduction of Invasive Alien Species ✓
- ☐ C Forest fires due to climate change
- ☐ D Human-wildlife conflict due to encroachment

✓ Answer: B — Invasive Alien Species disrupt existing ecosystem — species can't adapt to new arrivals

→ What is a Biodiversity Hotspot?

TWO CRITERIA (BOTH MUST BE MET)

- 1 Minimum **1,500 endemic vascular plant species**
- 2 Must have lost at least **70% of its original habitat**
Habitat = Natural home of species (like our home)

INDIA'S 4 BIODIVERSITY HOTSPOTS

-  **Himalayas**
-  **Western Ghats + Sri Lanka**
-  **Indo-Burma** (NE India + Myanmar)
-  **Sundaland** (Nicobar + SE Asia + Indonesia)

→ Invasive Alien Species (IAS)

Species that **do not originally belong** to a region but are introduced from outside.

CURRENT AFFAIRS

Example: Cheetahs brought from **Botswana / South Africa** to **Purna National Park** → started dying (can't adapt) AND threatened existing native species → could destroy entire ecosystem



② Medieval History — Mughal Empire

Q2. Whose administrative system later became the model for Emperor Akbar's consolidation of the Mughal Empire?

- ☐ A Babur
- ☐ B Hamagan
- ☒ C Sher Shah Suri ✓
- ☐ D Jahangir

✓ Answer: C — Sher Shah Suri (Afghan) → Akbar adopted his admin system | Sher Shah died 1545 at Kalinjar Fort

→ Babur (1526–1530)

Battle	Year	Against	Result
First Battle of Panipat	1526	Ibrahim Lodi	Babur won → Mughal Empire
Battle of Khanwa	1527	Rana Sanga (Sisodia Rajput, Mewar)	Babur won
Battle of Chanderi	—	Madini Rai	Babur won
Battle of Ghaghra	—	Afghans	Babur won

KEY FACTS

-  Ascended throne of Fergana at age **12** | Conquered Kabul in **1504**
-  Built **Aram Bagh** in Agra = **First Mughal Garden**
-  First buried in Agra → later shifted to Kabul (believed by Akbar/grandson)

→ Humayun (1530–1540 | 1555–56)

HUMAYUN'S DEFEATS

Lost to Sher Shah Suri at:

→ **Battle of Chausa (1539)**

→ **Battle of Kannauj / Bilgram (1540)**

Escaped to Iran (Safavid Dynasty)

HUMAYUN EXTRAS

Biography: **Humayunnama** by sister
Gulbadan Begum

Tomb: built by wife **Biga Begum / Haji Begum**

= First **Charbagh style architecture**

Died: fell from **library stairs**

→ Sher Shah Suri

Original name: **Farid Khan** | Afghan ruler | Died **1545** at **Kalinjar Fort** (injuries during attack)

→ **Akbar adopted Sher Shah's administrative system** ← That's why he is the correct answer!

→ Jahangir (1605–1627)

JAHANGIR FACTS

Wife: **Merun Nisa (Nur Jahan)**

She built: **Tomb of Itmad-ud-Daula**

= First use of **Pietra Dura** (gem/stone inlay in walls)

HALDIGHATI + MEWAR

Battle of Haldighati (1576) = Akbar's forces vs **Maharana Pratap**

Akbar didn't fight directly

Sisodia Rajputs of Mewar = accepted

Mughal suzerainty during **Jahangir's time**



③ Ecology — Important Terms

Q3. "Permaculture" — a term often seen in news — refers to what?

- ☐ A ~~Cropping and vegetation in permafrost regions~~
- ☐ B ~~Growth of organisms facilitated in a semi-solid medium~~
- ☐ C ~~A bioremediation technique to remove oil spills~~
- ☒ D **Development of sustainable agricultural ecosystems ✓**

✓ Answer: D — Permaculture = sustainable agri-ecosystems that mimic nature | NOT permafrost

→ Three Critical Bio-Terms

Term	Meaning	Example
Bioaccumulation	Toxins accumulate in tissues of one organism over time	Pesticides in our body from vegetables
Biomagnification	Toxin concentration increases up the food chain (trophic levels)	Grass→Chicken→Human → toxin level rises each step
Bioremediation	Using living organisms (bacteria, fungi) to remove pollutants	Cleaning oil spills using microbes

→ In-Situ vs Ex-Situ

IN-SITU CONSERVATION

Conservation at the **original habitat** of species

Ex: National Parks, Wildlife Sanctuaries, Biosphere Reserves

EX-SITU CONSERVATION

Conservation **outside original habitat**

Ex: Zoos, Botanical Gardens, Seed Banks, Gene Banks

→ Agriculture Terms (SSC Frequently Asked)

Term	Meaning
Viticulture	Cultivation of Grapes
Pisciculture	Farming of Fish
Horticulture	Gardening — Fruits & Vegetables Also linked to Golden Revolution
Apiculture	Beekeeping / Honey production Also Golden Revolution
Sericulture	Silk production
Silviculture	Forestry
Hydroponics	Growing plants in water (NO soil)



④ Ancient History — Harappan Civilization

Q4. Consider the following statements: 1. Seals of Harappan culture were made using locally found steatite. 2. These seals feature animal motifs and undeciphered script symbols. Which is/are correct?

- ☐ A Only 1
- ☐ B Only 2
- ☒ C Both 1 and 2 ✓
- ☐ D Neither 1 nor 2

✓ Answer: C — Both correct | Steatite from S. Rajasthan + N. Gujarat | Script = Pictographic, still undeciphered

HARAPPAN BASICS

Time Period: **2500–1750 BC**

Seals material: **Steatite (soft stone)**

Found in: S. Rajasthan + N. Gujarat

Script: **Pictographic** = still

UNDECIPHERED

Tamil Nadu govt offered **\$1 million** to decode it!

TRADE CONNECTIONS

Traded with **Mesopotamia** (Iraq civilization)

Harappan seals found there = proof of trade

Intermediate ports: **Dilmun** and **Magan**
Mesopotamians had a name for Harappans

→ Famous Finds from Mohenjo-daro

Object	Detail	Material / Technique
Pashupati Seal / Proto-Shiva Seal	Shows 5 animals: Buffalo, Elephant, Tiger, Rhino, Two Deer	Steatite
Priest King	17.5 cm statue	Steatite
Dancing Girl	Famous bronze statue	Lost Wax Technique

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⑤ Biology — Human Heart ❤️

Q5. Deoxygenated blood from the body enters the human heart through which of the following chambers?

☒ A Right Atrium ✓

☐ B Right Ventricle

☐ C Left Atrium

☐ D Left Ventricle

✓ Answer: A — Deoxygenated blood from body → Inferior/Superior Vena Cava → Right Atrium (+ SA node here)

→ Heart Chambers by Animal Species

Animal	Chambers	Type
Fish	2	Cold-blooded
Amphibians	3	Cold-blooded
Reptiles	3 (Exception: Crocodile = 4)	Cold-blooded
Birds	4	Warm-blooded
Mammals (incl. Humans)	4	Warm-blooded

→ Complete Blood Flow Path

- ☐ Step 1 Deoxygenated blood from body → **Inferior Vena Cava** (largest vein) + **Superior Vena Cava** from above
- ☐ Step 2 Blood enters **Right Atrium** → SA Node (natural pacemaker) here
- ☐ Step 3 Tricuspid Valve opens → **Right Ventricle**
- ☐ Step 4 Pulmonary Valve opens → **Pulmonary Artery** → Lungs (CO₂ out, O₂ in)
- ☐ Step 5 Oxygenated blood → **Pulmonary Vein** → **Left Atrium**
- ☐ Step 6 Mitral (Bicuspid) Valve opens → **Left Ventricle**
- ☐ Step 7 Aortic Valve opens → **Aorta** (largest artery) → Entire body

EXAM TRAP!

⚠ EXCEPTION: **Pulmonary ARTERY** carries **DEOXYGENATED** blood (arteries normally carry oxygenated)

⚠ EXCEPTION: **Pulmonary VEIN** carries **OXYGENATED** blood (veins normally carry deoxygenated)

KEY FACTS

- 🔑 **SA Node (Sinoatrial Node)** = **Natural Pacemaker** of heart | Located in Right Atrium
- 🔑 **Septum** = Wall dividing left and right sides of heart
- 🔑 **Pericardium** = Outer covering/layer of heart
- 🔑 **William Harvey** = First to discover blood circulation
- 🔑 Cockroach blood = **Colourless** (haemolymph, no haemoglobin)



⑥ Polity — Fundamental Duties

Q6. Which of the following is NOT a Fundamental Duty mentioned in the Indian Constitution?

- ☐ A ~~To respect ideals of the Constitution, national flag and anthem~~
- ☐ B ~~To uphold sovereignty, unity and integrity of India~~
- ☒ C **Separation of judiciary from executive ✓**
- ☐ D ~~To defend the country and render national service when called upon~~

✅ Answer: C — Separation of Judiciary from Executive = DPSP (Article 50, Part IV) NOT Fundamental Duty

FUNDAMENTAL DUTIES — BASICS

Added by: **42nd Amendment, 1976**
During: **Indira Gandhi's time**
Taken from: **USSR Constitution**
Added in: **Part IVA | Article 51A**
On recommendation: **Sardar Swaran Singh Committee**

KEY FEATURES

Nature: **Non-Justiciable** (courts can't enforce)
Only for: **Indian Citizens** (NOT foreigners)
Originally: **10 duties**
11th added by: **86th Amendment, 2002** (Vajpayee's time)
Justice Verma Committee = how to implement them

→ Separation of Judiciary from Executive = DPSP

DPSP vs FUNDAMENTAL DUTIES

Article 50 = Separation of Judiciary from Executive = **DPSP** (Part IV)

Taken from: **Constitution of Ireland** | Promotes welfare society

Also **Non-Justiciable** | Promotes social & economic democracy

→ All 11 Fundamental Duties — A to K Trick

Code	Duty
A	Respect ideals of Constitution, national flag and national anthem
B	Follow noble ideals of freedom struggle (Bapu + Bhagat Singh)
C	Uphold & protect Sovereignty, Unity, Integrity (3 things)
D	Defend the country & render national service when called upon
E	Promote harmony, brotherhood , dignity of women
F	Preserve rich heritage & composite culture (Fort = heritage)
G	Protect natural environment — Forests, Lakes, Rivers (Greenery)
H	Develop scientific temper, humanism
I	Safeguard public property, abjure violence
J	Strive towards excellence in all individual and collective activities
K (11th)	Parents/guardians educate children aged 6-14 years (86th Amendment 2002)

KEY FACTS

🔑 **86th Amendment 2002** did **THREE** things:

- 🔑 1. Added 11th Fundamental Duty (educate children 6-14)
- 🔑 2. Added **Article 21A** = Right to Education as Fundamental Right
- 🔑 3. Changed subject matter of **Article 45** = Early Childhood Care added



⑦ Geography — Peninsular Plateau

Q7. Consider the following physiographic features: Tors, Inselbergs, Block Mountains. Which of the above features are found in the Peninsular Plateau of India?

- ☐ A Only Tors
- ☐ B Only Tors and Inselbergs
- ☐ C Only Inselbergs and Block Mountains
- ☒ D All three ✓

✓ Answer: D — All three (Tors, Inselbergs, Block Mountains) are found in Peninsular Plateau

→ India's 6 Physiographic Divisions

1 The Himalayas	2 Northern Plains
3 Peninsular Plateau (LARGEST)	4 Indian Desert (W of Aravalli)
5 Coastal Plains	6 Islands (A&N + Lakshadweep)

→ Peninsular Plateau Features

Feature	Description
Tors	Rounded hills made of granite rock
Inselbergs	Residual isolated hills (like mounds/heaps in arid regions)
Block Mountains (Horst)	Land raised between two faults Sunken part = Graben (Rift Valley)

→ Sub-regions of Peninsular Plateau

CENTRAL HIGHLANDS (North of Narmada)

Mewar Plateau → Malwa Plateau → Bundelkhand → Baghelkhand → **Chota Nagpur Plateau**

Rajmahal Hills → Malda Gap → Meghalaya Plateau (Garo, Khasi, Jaintia) → Karbi Anglong Plateau

→ Western Ghats vs Eastern Ghats

Feature	Western Ghats	Eastern Ghats
Height	Higher (~1200 m avg)	Lower (~600 m avg)

Width	Narrower	Wider
Continuity	Continuous	Discontinuous
Coast Type	Submerging Coast	Emerging Coast
Coastal Plain	Narrow	Wide

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⑧ Modern History — Anglo-Mysore Wars

Q8. Who were the combined forces that attacked Tipu Sultan in 1792?

- ☐ A British and Marathas only
- ☐ B British and Nizam only
- ☒ C Marathas + Nizam + East India Company ✓
- ☐ D British, Marathas and French

✓ Answer: C — Third Anglo-Mysore War 1790–1792 | Treaty of Seringapatna | Tipu paid ₹3 crore + two sons as hostages

→ Four Anglo-Mysore Wars

War	Years	Key Event	Treaty
1st	1767–1769	Hyder Ali led; Mysore won	Treaty of Madras (1769)
2nd	1780–1784	Hyder Ali died of cancer → Tipu Sultan (Lion of Mysore) took over	Treaty of Mangalore
3rd	1790–1792	Marathas + Nizam + EIC attacked → Tipu humiliatingly defeated ₹3 crore indemnity + 2 sons as hostages	Treaty of Seringapatna
4th	1799	Tipu Sultan killed → Mysore Kingdom ended Wodeyar Dynasty restored	Subsidiary Alliance signed

→ Subsidiary Alliance (Lord Wellesley)

State	Year Accepted
Hyderabad (Nizam)	1798 — FIRST to accept
Mysore	1799

KEY FACTS

- 🔑 Ruler had to remove own army | Permanent **British Resident** in court
- 🔑 Cannot enter any treaty without British permission | British provide protection in return
- 🔑 Introduced by: **Lord Wellesley**



⑨ Geography — Karst Topography / Limestone Caves

Q9. Which of the following pairs is correctly matched with reference to limestone cave formation?

- ☐ A ~~Stalactite — rises from the cave floor~~
- ☒ B **Pillar — formed by union of stalactite and stalagmite ✓**
- ☐ C ~~Stalagmite — hangs tightly from ceiling~~
- ☐ D ~~Sinkhole — depositional landform~~

✓ Answer: B — Stalactite hangs FROM ceiling | Stalagmite rises FROM floor | Pillar = both meet

→ Karst Topography — Erosional Landforms

Named after **Karst region** near Mediterranean Sea where soft rocks (limestone, dolomite) are found. Groundwater acts on these → erosional + depositional landforms.

EROSIONAL LANDFORMS

- **Lapies** (irregular grooves on surface)
- **Limestone Pavements** (flat surface)
- **Sinkholes** (dissolved limestone depressions)
- **Dolines** (collapsed sinkholes)
- **Valleys** (sinkholes join together)
- **Caves** (underground hollow by erosion)

DEPOSITIONAL LANDFORMS (inside caves)

- 💧 **Stalactite** = hangs DOWN from ceiling
- 💧 **Stalagmite** = rises UP from floor
- 💧 **Pillar** = stalactite + stalagmite meet

🧠 MEMORY TRICK

🧠 **Memory Trick:** Stalactite = holds **TIGHT**ly to ceiling (tightly = tight = stalactite)

🧠 Open your **freezer/refrigerator** today! The icicle hanging from top = stalactite! The one rising from bottom = stalagmite!



⑩ Polity — New Criminal Laws (BNS) ⚖️

Q10. The BNS (Bharatiya Nyaya Sanhita) was introduced primarily as an effort to do what?

- ☐ A ~~Codify personal laws~~
- ☒ B **Modernize and decolonize India's criminal justice system ✓**
- ☐ C ~~New taxation policy of a trade agreement~~
- ☐ D ~~Introduce Uniform Civil Code~~

✅ Answer: B — BNS replaced IPC | 20 chapters, 358 sections | Zero FIR + 90-day investigation

→ Three Old Laws → Three New Laws

Old Law (British Era)	New Law (2023)
Indian Penal Code (IPC)	Bharatiya Nyaya Sanhita (BNS) — 20 chapters, 358 sections
CrPC	Bharatiya Nagarik Suraksha Sanhita (BNSS)
Indian Evidence Act	Bharatiya Sakshya Adhinियam (BSA)

🔑 KEY FACTS

- 🔑 Police must complete investigation within **90 days** of filing FIR (time-bound)
- 🔑 **Zero FIR** = FIR can be filed at ANY police station regardless of where crime occurred (asked in UPSC 2024!)



⑪ Art & Culture — Festivals of Kerala 🎉

Q11. The festival "Vet Tukda" celebrated in Kerala is associated with which religious community?

- ☐ A Sikh
- ☐ B Jain
- ☐ C Muslim
- ☒ D Christian ✓

✓ Answer: D — Vet Tukda = Christian community | Celebrated at Madre de Deus Church, Kerala

 MEMORY TRICK

 Memory Trick: "POORAN VASU DE BOATS ARE AT PEN"

Word	Festival	Details
POORAN	Thrissurpuram	Celebrated at Vadakumnatha Temple
VASU	Vishu	New Year festival of Kerala
DE	Deyam Festival	—
BOATS	Vallam Kali (Snake Boat Race)	Famous boat race of Kerala
ARE	Aarattu	At Padmanabhaswamy Temple (richest temple in India)
AT	Attukal Pongal	Pongal = "two boils" Celebrated for 4 days
PEN	Penkuni Festival	—
ONAM	Onam	Most important! Harvest festival Marks return of demon king Mahabali Sabarimala temple

 KEY FACTS

 **Makara Vilakku Festival** + **Nishagandhi Festival** = also Kerala

 **Vet Tukda Festival** = Christian community | Madre de Deus Church

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⑫ Environment — NAFCC & Water Harvesting

Q12. NAFCC (National Adaptation Fund for Climate Change) — what does it primarily aim at?

- ☐ A Improve microclimate resilience through community-led water harvesting in semi-arid zones
- ☒ B Climate Resilient Watershed Program ✓
- ☐ C Water-energy projects
- ☐ D Setting up solar panels in rural areas

✓ Answer: B — NAFCC launched 2015 | Government of India scheme | Climate Resilient Watershed Program

→ Traditional Water Harvesting Names

Region	Local Name
Jaisalmer, Rajasthan	Khadins and Johads
Himalayan Region	Kuls or Guls

♦ ♦ ♦

13 Modern History — Peasant Revolts

Q13. Which of the following revolts took place in Assam in 1906?

- ☒ A Revolt of Sangram Sangma ✓
- ☐ B Self-Respect Movement
- ☐ C Blue Rebellion
- ☐ D Kohli Revolt

✓ Answer: A — Revolt of Sangram Sangma = Assam 1906 | Elimination: Kohli=Gujarat; Self-Respect=South India; Blue Rebellion=Bengal

Revolt	Location	Year	Key Detail
Revolt of Sangram Sangma	Assam	1906	Correct answer for this question
Kohli Revolt	Gujarat & Maharashtra	1857–58	Kohli = fishermen community
Kol Revolt (different from Kohli!)	Jharkhand region	1831–32	Led by Budho Bhagat

Blue Rebellion (Indigo Revolt)	Nadia district, West Bengal	1859	Play Nil Darpan by Deen Bandhu Mitra / Translated by Michael Madhusudan Dutta
Self-Respect Movement	South India	1925	Started by EV Ramaswamy Naykar (Periyar)

EXAM TRAP!

⚠ **Kol ≠ Kohli** — Two DIFFERENT revolts! Don't confuse them in exam!

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14 Chemistry — CO₂ + Lime Water

Q14. When carbon dioxide is passed through lime water, which colour precipitate is obtained?

- ☐ A Yellow
- ☐ B Blue
- ☒ C White (Milky) ✓
- ☐ D Red

✓ Answer: C — CO₂ turns lime water milky white | CaCO₃ (Calcium Carbonate) = white precipitate

FORMULA



Lime water = Slaked Lime = Calcium Hydroxide = Ca(OH)₂

Type of reaction: **Combination Reaction** (two reactants → one product)

KEY FACTS

 Quick Lime (CaO) + Water → **Slaked Lime Ca(OH)₂** = Highly Exothermic Reaction
(lot of heat released)

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15 Geography — Rivers & Coal Deposits

Q15. Which river valley in the North Eastern Plateau is associated with coal deposits?

- ☐ A Tapti
- ☐ B Ganga
- ☐ C Brahmaputra
- ☒ D Damodar ✓

✓ Answer: D — Damodar flows in rift valley | Tributary of Hooghly | "Sorrow of Bengal" | Chota Nagpur = Ruhr of India

KEY FACTS

- 🔑 Damodar flows in a **rift valley**
- 🔑 Tributary of **Hooghly River** (which is a distributary of Ganga)
- 🔑 Called **"Sorrow of Bengal"** due to frequent floods
- 🔑 **Chota Nagpur Plateau** = Called **"Ruhr of India"** (richest in coal/minerals)



16 Geography — Demographic Transition Model

Q16. Assertion (A): Demographic statistics like birth & death rates rely on accurate reporting by households. Reason (R): Birth and death registration is mandatory and always accurately followed in all parts of India.

- ☐ A Both A and R true; R explains A
- ☐ B Both true; R does NOT explain A
- ☒ C A is true but R is FALSE ✓
- ☐ D A false, R true

✓ Answer: C — "always" + "all parts" = extreme keywords = FALSE | Registration NOT accurately followed everywhere in India

EXAM TRAP!

⚠ Extreme Keywords in Reason: **"always"** + **"accurately"** + **"all parts"** = THREE extreme keywords = Automatically WRONG statement!

→ Demographic Transition Model (DTM) — 4 Stages

Stage	Birth Rate	Death Rate	Population
Stage I	High	High	Stagnant

Stage 2	High	Falls sharply	Rapid increase
Stage 3	Falls sharply	Already low	Slowing down
Stage 4	Low	Low	Stable / Negative

→ India's Population History

- 1919–21 **Year of the Great Divide** — population growth was stagnant until this point
- 1921–51 Population increased due to better health facilities
- 1951–81 **Population Explosion Phase**
- After 1981 Population stabilizing
- 2011 Census **Total Fertility Rate (TFR) = 2.4** (range 2.1–2.9) | Currently falling to ~1.9

→ Population Pyramids

Shape	Growth Type
Wide Triangle (Pyramid)	Rapid Growth — more children → more reproduction
Narrow Pyramid	Slow Growth
Rectangle	Zero Growth
Bell-shaped (narrow base)	Negative Growth — fewer children, more elderly

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⑰ Physics — Thermodynamic Processes

Q17. What phenomenon is illustrated when a bicycle tyre bursts suddenly?

- ☒ A isothermal process
- ☒ B Adiabatic process ✓
- ☐ C isochoric process
- ☐ D isobaric process

✓ Answer: B — High pressure → sudden drop to low pressure → temperature drops sharply → feels very cold = Adiabatic

Process	What Remains Constant
Isothermal	Temperature

Adiabatic	No heat transfer in or out (but energy IS transferred) — tyre burst!
Isochoric	Volume
Isobaric	Pressure

CURRENT AFFAIRS

Real life examples of **Adiabatic process**: Tyre burst → finger feels very cold /
Continuously pressing deodorant gas can for 10–12 seconds → entire can cools down / Any sudden gas leak from high pressure to low pressure

♦ ♦ ♦

⑮ Polity — Fundamental Rights

Q18. Which of the following was removed from the list of Fundamental Rights?

- ☐ A Right to Education
- ☐ B Right to Constitutional Remedies
- ☒ C Right to Property ✓
- ☐ D Right to Freedom of Religion

✓ Answer: C — Right to Property removed by 44th Amendment / Now Legal Right under Article 300A (Part XII)

FUNDAMENTAL RIGHTS BASICS

Part: **Part III** | Articles: **12–35**

Originally: **7 FRs** | Now: **6 FRs**

Right to Property removed by: **44th Amendment**

From Articles: **31** and **19(1)(f)**

Now: Article **300A** = Legal Right only

IMPORTANT AMENDMENTS

Art. 31A & 31B → **1st Amendment**

Art. 31C → **25th Amendment**

Parliament can amend FRs → **24th Amendment**

Kesavananda Bharati Case = Basic Structure Doctrine

Minerva Mills (1980) = Balance between FRs and DPSPs

→ Article 20 — Three Key Protections

Protection	Meaning
No Ex-Post Facto Law	Punishment = as per law at time of crime, NOT a previous law

No Double Jeopardy	Cannot be punished <i>TWICE</i> for same crime
No Self-Incrimination	Cannot be forced to be witness against yourself

→ Right Against Exploitation — Articles 23 & 24

ARTICLE 23

Prohibits **human trafficking** and **forced labour** (bonded labour)

Forced labour = not paying wages or underpaying

ARTICLE 24

Prohibits **child labour** in hazardous work

Children (6–14) **CAN** do non-hazardous work during **non-school hours**

KEY FACTS

🔑 **Art 21** = Right to Life and Personal Liberty

🔑 **Art 21A** = Right to Education (86th Amendment 2002)

🔑 **Art 22** = Protection against arrest | Preventive detention max **3 months** without High Advisory Board

🔑 **Art 18** = Abolition of Titles



⑲ Ancient History — Charaka Samhita 📖

Q19. Charaka Samhita is a treatise on which of the following subjects?

☐ A Astronomy

☐ B Mathematics

☒ C Medicine (Ayurveda) ✓

☐ D Political Science

✅ Answer: C — Charaka Samhita = ancient Ayurvedic text | Originally compiled by Agnivasa | Charaka refined it during Kushan Dynasty

KEY FACTS

🔑 Originally compiled by **Agnivasa** (over very long period)

🔑 **Charaka** worked on it during the **Kushan Dynasty** period

🔑 Not compiled in a single time period — evolved over centuries



② Modern History — Charter Acts

Q20. The Charter Act of 1833 made the Governor General of Bengal into the Governor General of ___?

- ☐ A Madras Presidency
- ☐ B Bombay Presidency
- ☒ C India ✓
- ☐ D Bengal Province

✓ Answer: C — Charter Act 1833 → Governor General of Bengal → Governor General of India | First = William Bentinck

Act	Year	Key Contribution	First Person
Regulating Act	1773	First regulation of EIC Established Supreme Court Governor of Bengal → Governor General of Bengal	Warren Hastings
Charter Act	1813	₹1 lakh for Indian education Christian missionaries allowed EIC commercial monopoly ended (except tea + China)	—
Charter Act	1833	GG of Bengal → GG of India EIC became purely administrative Tea/China exception also removed Attempted open civil services	William Bentinck
Charter Act	1853	Actually introduced open competitive civil services	—
Indian Councils Act	1861	Started Decentralization — power restored to Bombay & Madras	—



Economics — Autonomous Investment & Multiplier

Q21. Assertion (A): Increase in autonomous investment leads to multiplier rise in national income. Reason (R): This happens because marginal propensity to save becomes less than one.

☒ A Both true; R is the correct explanation of A ✓

☐ B Both true; R does NOT explain A

☐ C A true; R false

☐ D A false; R true

✓ Answer: A — Both correct | Autonomous investment $\uparrow \rightarrow$ National Income $\uparrow$ by a multiple | Because $MPS < 1$ (money keeps re-circulating)

FORMULA

Total Consumption: $C = a + bY$

Where: **a** = Autonomous expenditure (income-independent) | **b** = MPC | **Y** = Income level

MPC + MPS = 1 (Total income = consumed + saved only)

MPC value: between **0 and 1** | MPS value: between **0 and 1**

KEY FACTS

🔑 **Autonomous Investment** = Part of investment **independent of income level**

🔑 **MPC** = Marginal Propensity to Consume = Change in Consumption / Change in Income

🔑 **MPS** = Marginal Propensity to Save = $1 - MPC$ | $MPS < 1$ = correct ✓

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Polity — Parliament Sessions (Article 85)

Q22. As per the Constitution, how many months shall NOT intervene between the last sitting of one session and the first sitting of the next session of Parliament?

☐ A 3 months

☒ B 6 months ✓

☐ C 4 months

☐ D 2 months

✓ Answer: B — Article 85 | Gap between 2 sessions $\leq$ 6 months $\rightarrow$ minimum 2 sessions per year

→ Three Sessions of Parliament

Session	Months	Notes
Budget Session	Feb – April/May	Longest session
Monsoon Session	July – August	Shortest session
Winter Session	December	Third session

→ Who Does What in Parliament?

Action	Who Does It
Summoning Parliament	President
Proroguing (ending session)	President
Dissolving Lok Sabha	President
Adjourning / Sine Die	Presiding Officer (Speaker)

🔑 KEY FACTS

- 🔑 **Quorum** (Art 100, Cl 3) = Minimum **1/10 (10%)** members must be present
- 🔑 **No-Confidence Motion** = If passed → Council of Ministers **MUST** resign | No reasons needed
- 🔑 **Censure Motion** = Needs approval of 50 members + reasons required | No mandatory resignation
- 🔑 First No-Confidence Motion against Nehru = brought by **JP Kripalani**

→ Quick Article Reference

Article	Content
79	Parliament
80	Rajya Sabha
81	Lok Sabha
82	Delimitation
83	Duration — Lok Sabha = 5 yrs RS members = 6 yrs, 1/3 retire every 2 yrs
84	Eligibility — Lok Sabha = 25 yrs Rajya Sabha = 30 yrs
85	Sessions of Parliament — gap ≤ 6 months
71	Election disputes of President/VP → settled by Supreme Court (exclusive jurisdiction)
72	Pardoning power of President

73	Extent of executive power of Union
74	Council of Ministers headed by PM advises President

◆ ◆ ◆

Current Affairs — Padma Awards 2025 🏆

Q23. Who received the Padma Shri in 2025 for contribution to the upliftment of the Musahar community?

- ☐ A ~~Sunita Williams~~
- ☒ B **Bhim Singh Bhosh ✓**
- ☐ C ~~Nirmala Devi~~
- ☐ D ~~Arundhati Bhattacharga~~

✓ Answer: B — Bhim Singh Bhosh | Musahar = Mahadalit community of Eastern Bihar | "Musa" = Mouse (they traditionally eat mice)

MUSAHAR COMMUNITY

Location: Primarily **Eastern Bihar**

Classification: **Mahadalit** community

Name origin: "**Musa**" = Mouse | They traditionally catch and eat mice

Several recent documentaries made about this community

◆ ◆ ◆

Economics — Import Substitution & Five-Year Plans 📈

Q24. What was the main rationale for the policy of import substitution during the first seven Five-Year Plans?

- ☐ A ~~To increase international trade~~
- ☐ B ~~To attract foreign investors~~
- ☒ C **To protect infant domestic industries ✓**
- ☐ D ~~To create export dependence~~

✓ Answer: C — Import Substitution = Inward-Looking Trade Strategy | Protect infant domestic industries from powerful foreign companies

IMPORT SUBSTITUTION

= NOT importing | Finding substitutes
within India

Also called: **Inward-Looking Trade Strategy**

Followed: Independence → 1990

Goal: Protect **infant domestic industries**

WHY IT FAILED

- Industries never got competition
- Efficiency never improved
- Balance of Payments crisis 1990
- Led to **LPG Reforms 1991**

→ LPG Reforms 1991

Letter	Reform	What Happened
L	Liberalisation	Ended License Raj Eased regulations
P	Privatisation	Brought in private players Disinvestment
G	Globalisation	Integrated Indian economy with world Foreign investment allowed

KEY FACTS

- Five-Year Plans concept borrowed from: **Soviet Union (USSR)**
- India also created **20-year long-term perspective plan**
- 7th Five-Year Plan = during **Rajiv Gandhi's** time as PM



Quick Revision Flash Table — All 24 Questions

Q.No	Topic	Answer
Q1	Biggest threat to Western Ghats species	Introduction of Invasive Alien Species (Option B)
Q2	Akbar's admin model from whom	Sher Shah Suri Original name Farid Khan Afghan
Q3	Permaculture meaning	Sustainable agricultural ecosystems (NOT permafrost)
Q4	Harappan seals — statements	Both correct Steatite + Pictographic undeciphered script
Q5	Deoxygenated blood enters heart at	Right Atrium via Inferior + Superior Vena Cava
Q6	NOT a Fundamental Duty	Separation of Judiciary = DPSP (Article 50)
Q7	Physiographic features of Peninsular Plateau	All three — Tors, Inselbergs, Block Mountains

Q8	Forces attacking Tipu in 1792	Marathas + Nizam + East India Company
Q9	Limestone cave formations	Pillar = stalactite + stalagmite meeting
Q10	Why BNS introduced	Modernize & decolonize criminal justice system
Q11	Vet Tukda festival community	Christian Madre de Deus Church, Kerala
Q12	NAFCC primary aim	Climate Resilient Watershed Program 2015
Q13	Assam revolt 1906	Revolt of Sangram Sangma
Q14	CO ₂ + lime water colour	White (Milky) CaCO ₃ precipitate
Q15	Coal deposits river valley — NE Plateau	Damodar River Sorrow of Bengal Chota Nagpur = Ruhr of India
Q16	Assertion-Reason: birth/death registration	A true, R false Extreme keywords: always + accurately + all parts
Q17	Tyre burst = which process	Adiabatic process (no heat transfer, energy transferred)
Q18	Removed from Fundamental Rights	Right to Property 44th Amendment Now Article 300A = Legal Right
Q19	Charaka Samhita subject	Medicine (Ayurveda) Kushan Dynasty period
Q20	Charter Act 1833 impact	GG of Bengal → GG of India First = William Bentinck
Q21	Autonomous investment assertion-reason	Both true; R explains A MPS < 1 causes multiplier effect
Q22	Parliament — gap between sessions	Max 6 months Article 85 Min 2 sessions per year
Q23	Padma Shri 2025 — Musahar community	Bhim Singh Bhosh Eastern Bihar Mahadalit
Q24	Import substitution rationale	Protect infant domestic industries Inward-Looking Strategy



Topper Tips — Exam Ke Liye Yaad Rakho!



EXAM TIP

- Pulmonary ARTERY** → carries DEOXYGENATED blood (exception!)
- Extreme keywords** in Reason = "always", "all", "never" = almost always FALSE
- Stalactite** = hangs from ceiling | Stalagmite = rises from floor



EXAM TIP

- Permaculture ≠ Permafrost!
Permaculture = sustainable farming
- 44th Amendment** = Right to Property removed | **42nd** = Fundamental Duties added
- Damodar = **Sorrow of Bengal** | Chota Nagpur = **Ruhr of India**

4. **Kol ≠ Kohli** — Two completely different revolts! Don't confuse!

8. Zero FIR = file at **ANY police station** regardless of where crime happened



END OF BLADE SERIES LECTURE 11 NOTES

All 24 Questions Covered  | Practice karo, Mock do, Top karo! 

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